

FORT SIMPSON, NT, Canada

WMO: 713650

Lat: 61.7587N

Lon: 121.2301W

Elev: 551

StdP: 14.41

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.2	-34.3	-43.0	0.5	-35.5	-37.6	0.7	-31.2	19.4	1.0	17.8	-1.1	0.8	320	0.667

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	83.2	63.0	80.1	61.5	76.9	60.2	65.5	78.3	63.8	75.9	62.2	73.3	6.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.9	81.6	68.6	59.1	76.3	67.7	57.4	71.7	66.7	30.7	78.4	29.4	76.3	28.2	73.6	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	15.9	13.6	DB	-44.1	89.2	5.7	3.3	-48.2	91.6	-51.5	93.5	-54.7	95.4	-58.8	97.9
			WB	-41.0	68.8	4.3	2.1	-44.1	70.3	-46.6	71.5	-49.0	72.6	-52.2	74.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.0	-10.3	-2.9	8.4	30.7	48.7	60.0	63.8	59.0	47.4	30.0	5.9	-6.7	
	DBStd	28.85	14.46	12.35	14.89	11.84	9.64	7.06	5.83	7.01	8.02	8.48	11.89	13.30	
	HDD50	9223	1870	1481	1289	582	138	11	0	11	141	620	1323	1757	
	HDD65	13615	2335	1901	1754	1029	509	179	92	209	528	1084	1773	2222	
	CDD50	1190	0	0	0	2	97	309	429	289	63	1	0	0	
	CDD65	109	0	0	0	0	3	27	56	22	1	0	0	0	
	CDH74	1347	0	0	0	0	86	355	609	280	17	0	0	0	
	CDH80	289	0	0	0	0	11	75	137	64	2	0	0	0	
Wind	WSAvg	5.3	4.4	4.9	6.1	6.1	6.3	5.7	5.3	5.4	5.4	5.1	4.8	4.4	
Precipitation	PrecAvg	14.8	0.7	0.7	0.7	0.7	1.2	1.9	2.3	2.2	1.2	1.5	0.9	0.7	
	PrecMax	24.9	1.6	2.2	2.0	2.6	3.9	7.0	5.2	4.6	3.7	6.1	1.9	2.2	
	PrecMin	9.7	0.0	0.2	0.0	0.0	0.1	0.3	0.5	0.4	0.3	0.1	0.2	0.3	
	PrecStd	3.1	0.4	0.5	0.5	0.6	0.9	1.3	1.1	1.2	0.7	1.0	0.5	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.4	38.8	50.5	64.9	81.2	86.7	88.0	86.5	75.5	59.1	40.8	35.2	
		MCWB	32.3	33.1	39.3	48.1	57.1	62.7	65.6	65.7	58.3	47.2	34.3	30.9	
	2%	DB	22.3	25.2	43.3	58.5	76.1	81.8	83.6	80.8	70.0	50.6	29.7	21.9	
		MCWB	21.3	22.8	35.1	44.4	55.5	60.9	63.9	62.8	56.0	43.1	28.4	20.7	
	5%	DB	13.9	19.1	37.0	54.1	71.4	78.0	80.4	76.7	65.5	45.1	25.4	17.3	
		MCWB	13.2	17.7	31.7	41.9	53.8	59.2	62.9	61.2	53.6	40.4	24.4	16.5	
	10%	DB	7.8	13.7	30.5	49.3	66.3	74.5	77.1	72.8	61.3	40.8	21.4	12.0	
		MCWB	7.1	12.6	27.0	39.4	50.6	58.0	61.7	59.5	52.0	37.6	20.5	11.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.3	33.5	39.7	49.1	59.6	65.6	68.2	67.2	60.9	48.5	34.5	31.3
			MCDB	37.8	38.5	49.5	63.8	76.4	78.8	82.1	82.3	72.7	56.5	38.9	34.9
2%		WB	21.3	23.1	35.2	44.9	56.4	63.3	66.2	64.6	57.3	43.9	28.4	20.9	
		MCDB	22.6	25.1	42.5	57.6	73.3	76.1	79.0	77.2	67.2	49.6	29.6	21.8	
5%		WB	13.5	17.7	31.6	42.3	54.1	61.4	64.6	62.4	54.7	40.7	24.4	16.5	
		MCDB	14.2	19.3	36.9	53.2	70.2	73.6	76.8	73.6	63.1	44.9	25.4	17.3	
10%		WB	7.5	12.7	27.2	39.6	51.4	59.5	63.1	60.5	52.6	37.8	20.6	11.4	
		MCDB	8.1	13.7	30.6	48.9	65.5	71.2	74.4	70.5	60.2	40.6	21.4	12.0	
Mean Daily Temperature Range	5% DB	MDBR	13.7	17.2	21.2	20.9	22.3	21.4	20.7	20.2	18.7	11.5	12.3	13.3	
		MCDBR	20.9	23.2	25.4	26.8	28.9	26.7	26.5	26.8	26.4	20.4	13.1	17.8	
		MCWBR	19.5	21.2	18.3	15.6	14.1	11.7	11.8	12.8	14.4	13.9	11.9	16.5	
	5% WB	MCDBR	21.0	23.3	24.9	25.9	27.3	22.7	23.4	23.8	23.9	19.0	13.1	18.0	
		MCWBR	19.6	21.3	18.4	15.3	13.8	10.8	11.2	12.3	13.9	13.3	11.9	16.7	
Clear-Sky Solar Irradiance	taub	0.193	0.246	0.267	0.333	0.344	0.354	0.380	0.354	0.308	0.273	0.199	0.157		
	taud	2.111	2.260	2.264	2.198	2.427	2.443	2.390	2.464	2.582	2.521	2.140	1.958		
	Ebn at Noon	177	238	273	270	278	277	265	263	260	231	170	130		
	Edh at Noon	10	19	27	36	32	32	33	28	21	15	10	7		
All-Sky Solar Radiation	RadAvg	108	384	956	1575	1805	1870	1713	1335	843	364	128	53		
	RadStd	5	21	43	75	127	157	113	94	85	32	15	3		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	+3.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page